Maximum Estimated Latency in a Tree

Problem statement

You are working on a compute cluster at NVIDIA. The cluster's interconnect is structured as a tree: **N** compute nodes connected by **N − 1** bidirectional links, with no cycles. Each link has a positive transmission latency measured in microseconds.

A message traveling between two different nodes follows the unique path connecting them. For capacity planning, the estimated latency includes the normal transmission time plus one additional transmission over the slowest link on that path.

For a path containing link latencies w1, w2, ..., wk, define:

```
Estimated latency = (w1 + w2 + ... + wk) + max(w1, w2, ..., wk)
```

The additional penalty is applied **once per path**, even if several links tie for the largest latency. It applies only to a link on the selected path. It does not create a detour or add any other link traversal.

For example, a path with link latencies 4, 2, 5 has estimated latency:

```
4 + 2 + 5 + max(4, 2, 5) = 11 + 5 = 16 microseconds
```

Task

Given the cluster topology, compute the **maximum estimated latency between any pair of distinct nodes**. If the cluster contains only one node, output 0.

Input format

```
N
u1 v1 w1
u2 v2 w2
...
u(N-1) v(N-1) w(N-1)
```

- The first line contains the integer N, the number of compute nodes. Nodes are numbered from 1 to N.
- Each of the next N − 1 lines contains three integers u, v and w, describing a bidirectional link between nodes u and v with latency w microseconds.
- Each input contains one tree. There is no leading test-case count.

Output format

Print a single integer: the maximum estimated latency, in microseconds.

Constraints

- $1 \le N \le 200{,}000$
- $1 \le u, v \le N$, with $u \ne v$
- $1 \le w \le 1{,}000{,}000{,}000$
- The input always forms a valid tree: it is connected and has no cycles.

Sample case 0

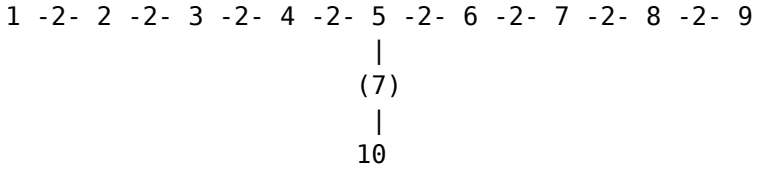
Input:

```
10
1 2 2
2 3 2
3 4 2
4 5 2
5 6 2
6 7 2
7 8 2
8 9 2
5 10 7
```

Output:

```
22
```

Explanation: nodes 1 through 9 form a horizontal chain where every link has latency 2. A vertical link between node 5 and node 10 has latency 7.



Some paths and their estimated latencies:

Endpoints	Link latencies on the path	Normal latency	Additional penalty	Estimated latency
1 and 9	2, 2, 2, 2, 2, 2, 2, 2	16	2	18
10 and 1	7, 2, 2, 2, 2	15	7	22
10 and 9	7, 2, 2, 2, 2	15	7	22

The maximum estimated latency is 22 microseconds.

Sample case 1

Input:

```
2
1 2 7
```

Output:

```
14
```

Explanation: there is only one possible pair of distinct nodes. Its path contains one link with latency 7, so the estimated latency is 7 + 7 = 14 microseconds.

Notes

- The objective includes both the sum and the largest link latency on the selected path. Maximizing only the sum is not sufficient.

- Use 64-bit integer arithmetic for accumulated latencies and the answer. The answer can be as large as 200,000,000,000,000.
- An O(N) solution exists. Solutions that enumerate all pairs of nodes have quadratic complexity and are not suitable for the largest inputs.
- The tree can be a chain containing 200,000 nodes. Account for traversal depth when implementing your solution.

Transcribed from photos of the screen. The opening sentence was cut off and is reconstructed, and the Sample 0 input lines are rebuilt from the explanation, so the original edge order may differ.